many more massless spin1⁄2 vibrational patterns than there are particles in Table 12.1. Moreover, the list of spin-½ vibrational patterns shows no trace of any repetitive groupings like the family structure of Table 12.1. With a less cursory inspection, then, it seems increasingly difficult to see how string vibrations will align with the known particle species.

Thus, by the mid-1980s, while there were reasons to be excited about superstring theory, there were also reasons to be skeptical. Undeniably, superstring theory presented a bold step toward unification. By providing the first consistent approach for merging gravity and quantum mechanics, it did for physics what Roger Bannister did for the four-minute mile: it showed the seemingly impossible to be possible. Superstring theory established definitively that we could break through the seemingly impenetrable barrier separating the two pillars of twentieth-century physics.

Yet, in trying to go further and show that superstring theory could explain the detailed features of matter and nature's forces, physicists encountered difficulties. This led the skeptics to proclaim that superstring theory, despite all its potential for unification, was merely a mathematical structure with no direct relevance for the physical universe.

Even with the problems just discussed, at the top of the skeptics' list of superstring theory's shortcomings was a feature I've yet to introduce. Superstring theory does indeed provide a successful merger of gravity and quantum mechanics, one that is free of the mathematical inconsistencies that plagued all previous attempts. However, strange as it may sound, in the early years after its discovery, physicists found that the equations of superstring theory do *not* have these enviable properties if the universe has three spatial dimensions. Instead, the equations of superstring theory are mathematically consistent only if the universe has *nine* spatial dimensions, or, including the time dimension, they work only in a universe with *ten* spacetime dimensions!

In comparison to this bizarre-sounding claim, the difficulty in making a detailed alignment between string vibrational patterns and known particle species seems like a secondary issue. Superstring theory requires the existence of *six* dimensions of space that no one has ever seen. That's not a fine point—*that's* a problem.